# INTERIM PROJECT REPORT

# Predictive Maintenance

Business Problem, Data Registration, Exploratory Data Analysis, Data Preparation, Model Building and Experiment Tracking

**Learner:** Akram Shazzad Shaik

**Dataset:** engine_data.csv

**Report type:** Interim Project Report

## Executive Summary

This project evaluates whether six engine measurements contain information useful for classifying the recorded *Engine Condition* labels. The dataset contains 19,535 records and seven columns. Validation reported no missing cells and no exact duplicate rows.

Four candidate classifiers were compared using stratified three-fold cross-validation on the training partition. AdaBoost achieved the highest mean cross-validation weighted F1-score (0.6462) and was selected. On the held-out test set, it achieved 65.42% accuracy, 0.6385 weighted F1-score, and 0.6846 ROC-AUC.

**Interpretation caveat:** The supplied materials do not define whether labels 0 and 1 represent failure, normal operation, or another state. These results demonstrate classification performance on this dataset; they do not establish advance failure prediction or production readiness.

### Report scope

This interim report covers data registration, EDA, data preparation, model tuning, experiment tracking, and preliminary business implications. Deployment and quantified financial benefits are outside the current scope.

# 1. Business Problem Understanding

**Business context:** Unplanned equipment downtime may increase maintenance costs and disrupt reliability. Telemetry analytics may help identify patterns that warrant investigation when the target label and operating context are defined.

**Problem statement:** Determine whether engine RPM, lubricating-oil pressure, fuel pressure, coolant pressure, lubricating-oil temperature, and coolant temperature contain information useful for classifying the recorded Engine Condition label.

**Objectives:** Validate the data; explore target balance, feature distributions, and relationships; prepare a reproducible train/test split; compare tuned classification models; and document findings and limitations.

# 2. Data Registration and Validation

**Methodology:** The registered dataset is `data/engine_data.csv`. Validation checks the expected columns, non-empty input, missing values, binary target values, row/column counts, duplicate rows, and target distribution.

Table 1. Dataset validation summary

| Data check | Result |
|---|---|
| Rows | 19,535 |
| Columns | 7 |
| Predictors | 6 |
| Missing cells | 0 |
| Duplicate rows | 0 |
| Target labels | 0 and 1 |

Table 2. Dataset fields and roles

| Column | Role / interpretation |
|---|---|
| Engine rpm | Numeric engine measurement; units and acceptable ranges require domain confirmation. |
| Lub oil pressure | Numeric engine measurement; units and acceptable ranges require domain confirmation. |
| Fuel pressure | Numeric engine measurement; units and acceptable ranges require domain confirmation. |
| Coolant pressure | Numeric engine measurement; units and acceptable ranges require domain confirmation. |
| lub oil temp | Numeric engine measurement; units and acceptable ranges require domain confirmation. |
| Coolant temp | Numeric engine measurement; units and acceptable ranges require domain confirmation. |
| Engine Condition | Target label; operational meaning of 0 and 1 needs confirmation. |

The initial checks support proceeding to EDA but do not prove that sensor values are physically valid. Sensor units, acceptable ranges, and the label-generation process should be confirmed with a domain expert.

# 3. Exploratory Data Analysis (EDA)

**Methodology:** The EDA in the executed notebook covers data overview, univariate analysis, bivariate analysis, and multivariate analysis. It uses target counts and proportions, descriptive statistics, feature histograms and box plots, class-wise comparisons, a correlation heatmap, and target-coloured pairwise scatter plots.

**Business interpretation:** Findings are descriptive and do not establish causation or confirm actual engine-failure behaviour. The target-label meanings and sensor ranges should be validated with engine-domain experts before operational decisions.

## 3.1 Target distribution

Figure 1. Distribution of Engine Condition labels (from the executed notebook).

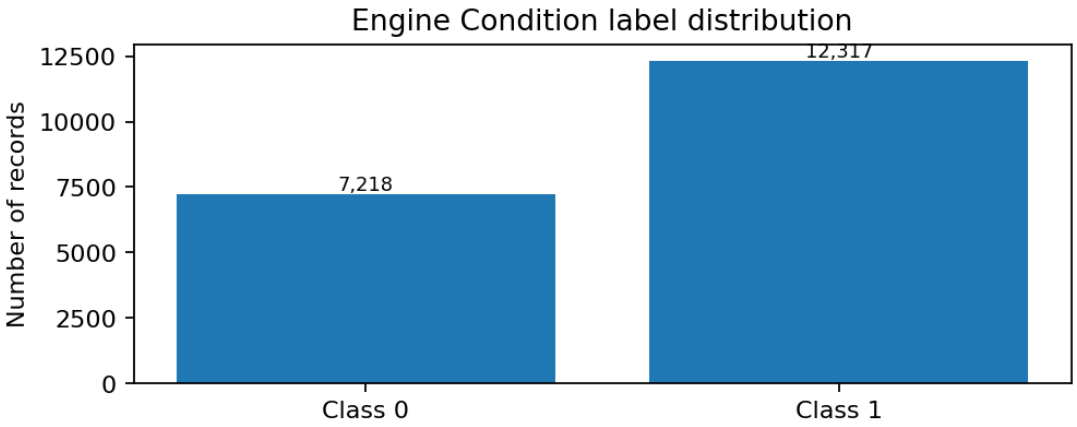



Table 3. Target class counts and proportions

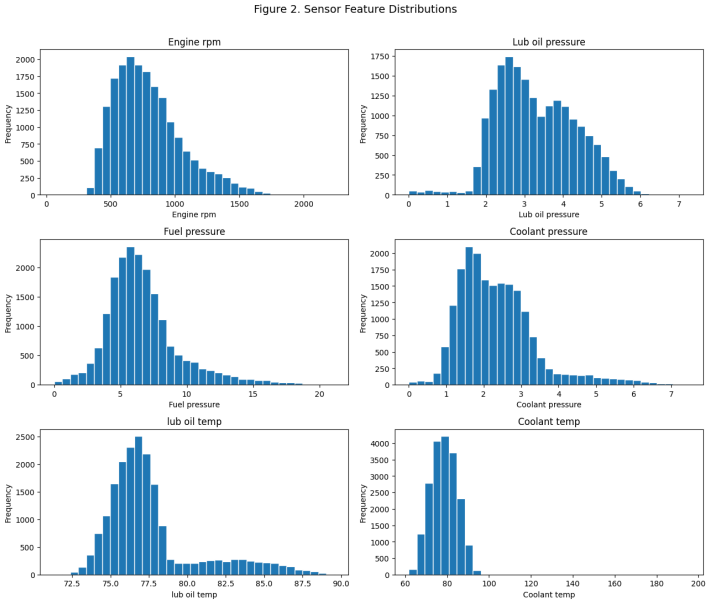
| Class | Count | Share |
|---|---|---|
| 0 | 7,218 | 36.95% |
| 1 | 12,317 | 63.05% |

Class 1 accounts for 63.05% of records and Class 0 for 36.95%. The distribution is uneven, so evaluation should include per-class measures and comparison with a baseline rather than accuracy alone.

## 3.2 Univariate analysis: sensor distributions

Histograms examine feature distributions and spread; box plots screen for potentially unusual values. Extreme observations are not automatically treated as errors or removed.

Figure 2. Sensor feature distributions from the executed notebook.



# 3. Exploratory Data Analysis (continued)

## 3.2 Univariate analysis: box plots and summary statistics

Figure 3. Sensor feature box plots for outlier screening (executed notebook).

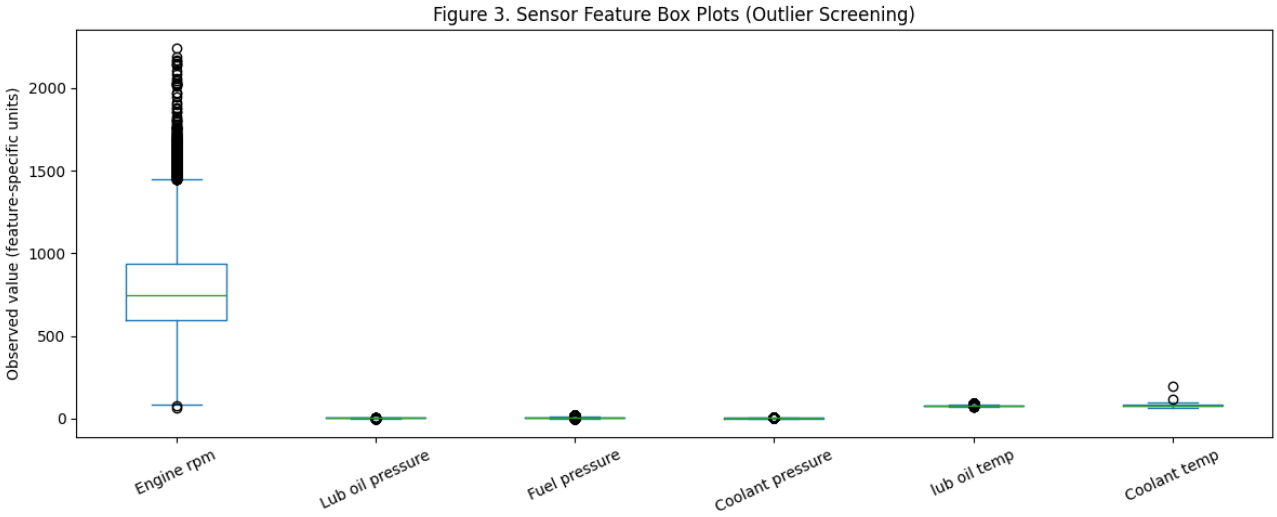


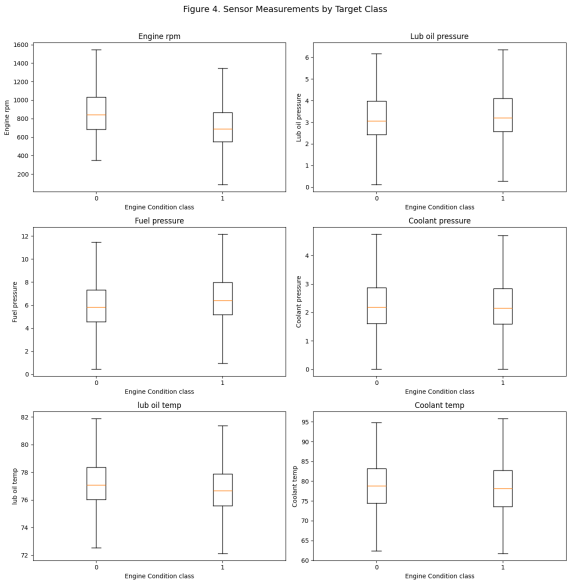

Table 4. Summary statistics for sensor features

| Feature | Mean | Std. dev. | Min | Median | Max |
|---|---|---|---|---|---|
| Engine rpm | 791.24 | 267.61 | 61.00 | 746.00 | 2239.00 |
| Lub oil pressure | 3.30 | 1.02 | 0.00 | 3.16 | 7.27 |
| Fuel pressure | 6.66 | 2.76 | 0.00 | 6.20 | 21.14 |
| Coolant pressure | 2.34 | 1.04 | 0.00 | 2.17 | 7.48 |
| lub oil temp | 77.64 | 3.11 | 71.32 | 76.82 | 89.58 |
| Coolant temp | 78.43 | 6.21 | 61.67 | 78.35 | 195.53 |

The reported maximum coolant temperature (195.53) is much higher than its median (78.35). This warrants checking against source documentation and sensor specifications; it should not be removed automatically without domain validation.

## 3.3 Bivariate analysis: feature distributions by class

Figure 4. Sensor measurements by target class (box plots from the executed notebook).




Class-wise mean comparisons show the largest absolute mean difference for engine RPM (148.698), followed by fuel pressure (0.665), lubricating-oil temperature (0.604), and coolant temperature (0.596). These differences may indicate

predictive signal but do not establish causation or define a particular engine condition.

# 3. Exploratory Data Analysis (continued)

## 3.4 Multivariate analysis: correlation heatmap

Figure 5. Correlation matrix for sensor features and Engine Condition (executed notebook).

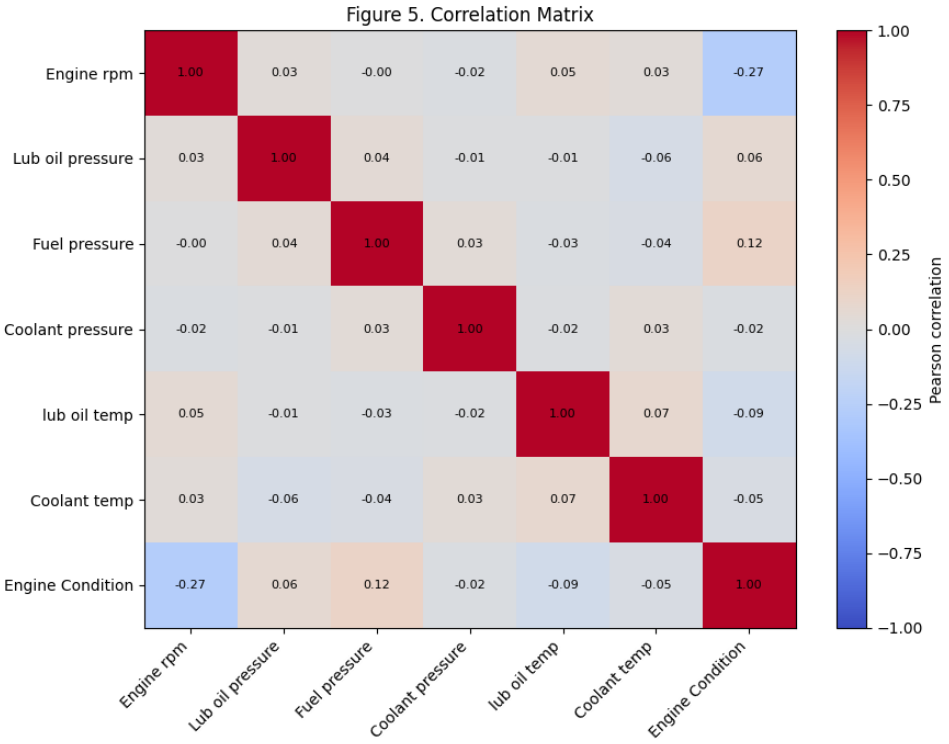


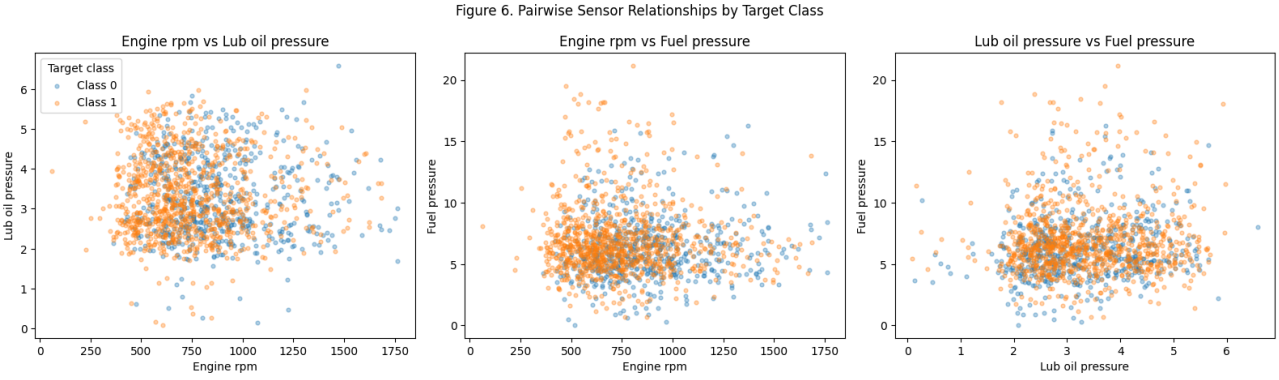

Table 5. Pearson correlation of each feature with the target

| Feature | Correlation with target |
|---|---|
| Engine rpm | -0.268 |
| Fuel pressure | 0.116 |
| lub oil temp | -0.094 |
| Lub oil pressure | 0.061 |
| Coolant temp | -0.046 |
| Coolant pressure | -0.024 |

Engine RPM has the largest absolute linear correlation with the target (-0.268), which is modest. Correlation may miss nonlinear relationships and interactions, and the direction cannot be translated into failure risk until label meanings are confirmed.

## 3.4 Multivariate analysis: pairwise sensor relationships

Figure 6. Pairwise sensor relationships coloured by target class (sample of up to 1,500 records).



## 3.5 EDA findings and business implications

• Class distribution is uneven: Class 1 represents 63.05% and Class 0 represents 36.95%.

• Engine RPM has the largest absolute class-wise mean difference and the largest absolute feature-to-target linear correlation.

• Feature distributions overlap; no single sensor should be treated as a stand-alone maintenance rule from these plots.

• Sensor units, unusual readings, label definitions, and the relationship between records and engines/time need domain validation.

# 4. Data Preparation

**Methodology:** A stratified 80/20 split was generated with fixed random seed 42. Stratification approximately preserves target-class proportions. The split files are saved as workflow artifacts and are present in the repository under `artifacts/splits/`.

Table 6. Train/test split

| Split | Rows | Purpose |
|---|---|---|
| Training set | 15,628 | Fit candidate models and perform cross-validation tuning. |
| Test set | 3,907 | Final evaluation after model selection. |

No scaling was required for the tree-based models evaluated. If records repeat by asset or time, an asset-aware or time-aware split may better reflect real-world generalization; this should be checked with the data owner.

# 5. Model Building and Experiment Tracking

**Methodology:** A majority-class DummyClassifier provides a baseline. Decision Tree, Random Forest, AdaBoost, and Gradient Boosting were tuned using GridSearchCV with stratified three-fold cross-validation. Mean weighted F1 was the primary selection metric. The held-out test set was not used to select the winning model.

Table 7. Cross-validation model comparison

| Candidate model | Best CV weighted F1 | Best parameters reported |
|---|---|---|
| AdaBoost | 0.6462 | learning_rate=1.0; n_estimators=20 |
| Gradient Boosting | 0.6375 | learning_rate=0.1; max_depth=2; n_estimators=40 |
| Random Forest | 0.6350 | n_estimators=40; max_depth=None; min_samples_leaf=1 |
| Decision Tree | 0.6078 | criterion=gini; max_depth=8; min_samples_leaf=1 |

AdaBoost was selected because it achieved the highest mean cross-validation weighted F1-score (0.6462).

# 5. Model Evaluation

Table 8. Held-out test evaluation results

| Model | Accuracy | Precision (weighted) | Recall (weighted) | F1 (weighted) | F1 (macro) | ROC-AUC |
|---|---|---|---|---|---|---|
| AdaBoost | 0.6542 | 0.6384 | 0.6542 | 0.6385 | 0.6002 | 0.6846 |
| Gradient Boosting | 0.6660 | 0.6499 | 0.6660 | 0.6336 | 0.5861 | 0.6981 |
| Random Forest | 0.6424 | 0.6254 | 0.6424 | 0.6266 | 0.5873 | 0.6645 |
| Decision Tree | 0.6063 | 0.6449 | 0.6063 | 0.6131 | 0.6009 | 0.6537 |
| Dummy baseline | 0.6304 | 0.3974 | 0.6304 | 0.4875 | 0.3867 | N/A |

Gradient Boosting has the highest reported test accuracy (66.60%) and ROC-AUC (0.6981), but AdaBoost remains selected because the predefined selection rule uses training-only cross-validation weighted F1. Test scores are not used to retroactively change model selection.

## 5.1 Selected model: class-level results

| Class / average | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Class 0 | 0.5453 | 0.3878 | 0.4533 | 1,444 |
| Class 1 | 0.6931 | 0.8104 | 0.7471 | 2,463 |
| Macro average | 0.6192 | 0.5991 | 0.6002 | 3,907 |
| Weighted average | 0.6384 | 0.6542 | 0.6385 | 3,907 |

## 5.2 Model evaluation plots

Figure 7. Confusion matrix for the selected AdaBoost model.

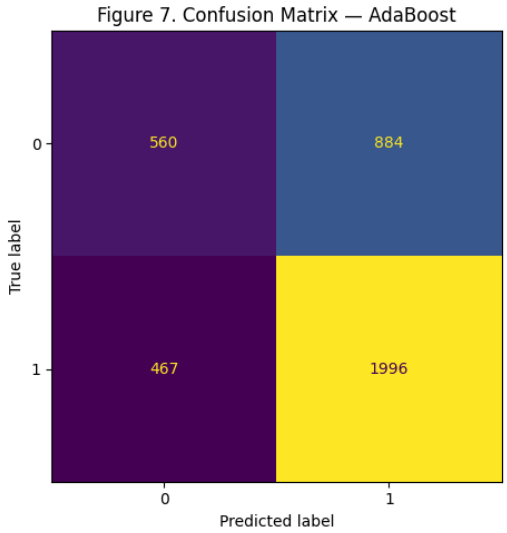
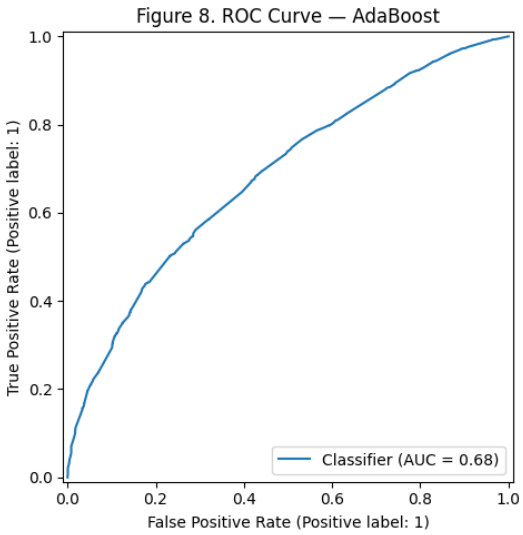



Figure 8. ROC curve for the selected AdaBoost model.




AdaBoost recalls Class 1 more often than Class 0. The meaning of each class must be confirmed before assigning a cost to either error type.

## 5.3 Feature Importance

**Methodology:** The selected estimator exposes `feature_importances_`; the notebook ranks sensor features by the fitted model's internal importance scores. These scores are not causal effects and may be influenced by correlations among predictors.

Figure 9. AdaBoost feature importance from the executed notebook.

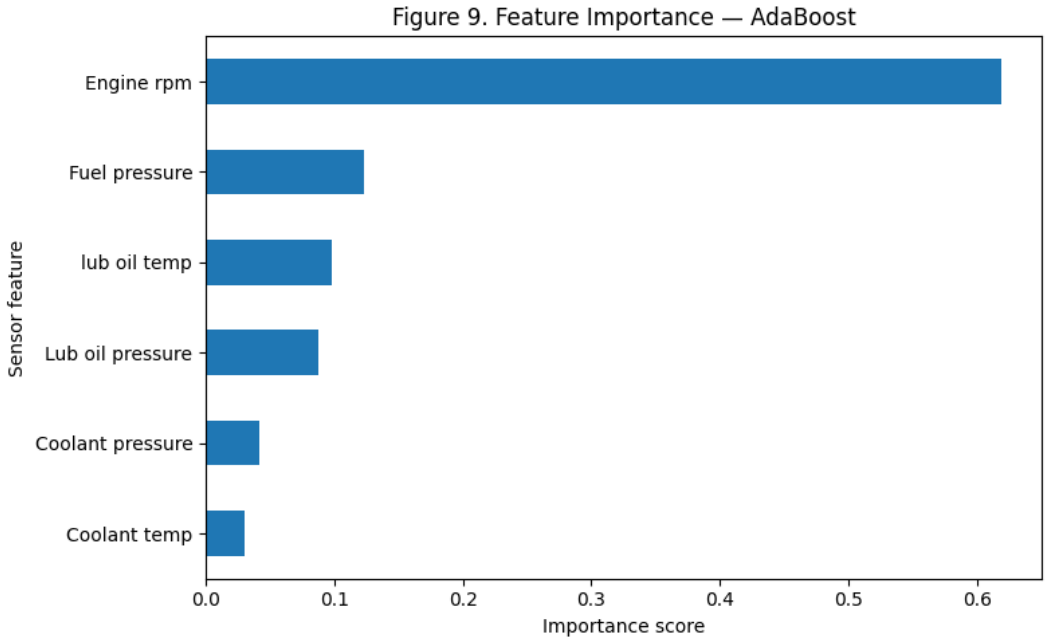



Table 9. Feature importance

| Feature | Importance |
|---|---|
| Engine rpm | 0.6194 |
| Fuel pressure | 0.1233 |
| lub oil temp | 0.0979 |
| Lub oil pressure | 0.0878 |
| Coolant pressure | 0.0417 |
| Coolant temp | 0.0300 |

# 6. Interim Findings and Business Implications

• **Data quality:** Schema validation found the expected columns, no missing cells, and no exact duplicate rows. Domain validation is still needed for unusual sensor readings.

• **Predictive signal:** EDA identifies class-wise distribution differences; model validation tests whether features are useful together.

• **Operational trade-off:** Review false negatives and false positives. If missed failures are more costly than false alarms, stakeholders should define the operational objective and threshold.

• **Deployment readiness:** A saved model is not proof of production readiness. Confirm labels, validate on representative future data, monitor drift, and agree alert thresholds.

• **Reproducibility:** Dataset, split files, experiment log, evaluation outputs, figures, model, and metadata should be verified in project folders.

# 7. Limitations and Questions for Mentor / Data Owner

• What do labels 0 and 1 mean operationally, and how were they assigned?

• What are the sensor units and acceptable operating ranges?

• Are records repeated by engine/asset or time, and can timestamps or asset IDs be provided?

• Is the objective to classify current state or predict failure a defined time before it occurs?

• Which error is more costly: missed maintenance risk or unnecessary inspection?

• What business-defined performance threshold and external validation set should be used?

The dataset appears to be a static labelled dataset in the reviewed columns. Model results are preliminary and should not be used alone to trigger maintenance decisions.

# 8. Conclusion and Next Steps

The project establishes a repeatable workflow from data validation through EDA, train/test preparation, model tuning, and experiment tracking. AdaBoost was selected using training-only cross-validation, with mean weighted F1 of 0.6462; held-out test accuracy was 65.42% and ROC-AUC was 0.6846.

Next steps are to confirm target-label definitions and sensor ranges, review confusion-matrix and class-level errors with the mentor, and evaluate the selected model on appropriately separated data before discussing deployment or financial benefits.

# Appendix A. Reproducibility and Workflow Evidence

The executed notebook and HTML export provide code and execution outputs. Representative repository paths include `data/engine_data.csv`, `src/validate_data.py`, `src/prepare_data.py`, `src/train_model.py`, `artifacts/splits/train.csv`, `artifacts/splits/test.csv`, and `artifacts/model/`.

The GitHub Actions workflow was observed to complete both jobs successfully: data validation/split creation and model tuning/evaluation. This is pipeline execution evidence, not proof of production readiness.

# Appendix B. Dataset Provenance

Dataset: `engine_data.csv`, the file supplied with the project materials. The original public dataset URL or publication details were not included in the reviewed materials. Add the original source and access date if known; do not cite an unverified source.

EDA figures in this revised report were taken from the uploaded executed HTML notebook. Summary statistics, class-wise comparisons, correlations, model metrics, and feature-importance values are aligned to the notebook outputs.